The CTC Algorithm — Solutions Manual

manim-concepts

Chapter 1

Counting rules

Practice problems

Problem 1.1. A per-frame labeller writes one of 3 symbols at each of 4 frames, independently. How many distinct label sequences can it write?

Solution. Four independent stages of 3 options each: $3 \times 3 \times 3 \times 3 = 3^4 = 81$. (This is the raw path space the next chapter's model writes over — the number returns almost immediately.)

Problem 1.2. Eight runners; how many distinct podiums (gold, silver, bronze) are possible?

Solution. $P_3^8 = 8 \times 7 \times 6 = 336$. In factorial form, $8!/5! = 336$: write the full ordering count and cancel the five unused slots' tail.

Problem 1.3. From the same eight runners, how many three-person relay *teams* (no roles) can be formed? Check your answer against the previous problem using the overcount argument.

Solution. Each unordered team of 3 appears $3! = 6$ times among the 336 podiums, so $\binom{8}{3} = 336/6 = 56$. Equivalently $8!/(3!\,5!) = 56$. The check is the definition: $C = P/r!$.

Problem 1.4. How many distinct arrangements does the word SEEDED have?

Solution. Six letters with repeats $E \times 3$, $D \times 2$, $S \times 1$:

$$\frac{6!}{3!\,2!\,1!} = \frac{720}{12} = 60.$$

The partition rule — the divide-out move applied per block of equal letters.

Problem 1.5. [stretch] A walker on a grid moves from the bottom-left corner to the top-right corner of a 2-row-by-4-column lattice of steps, moving only right or up — 4 rights and 2 ups in some order. How many routes are there?

Solution. A route is an arrangement of the multiset $\{R, R, R, R, U, U\}$: $\binom{6}{2} = \frac{6!}{4!\,2!} = 15$ routes — choosing which 2 of the 6 steps go up. Keep this number; a lattice with exactly this count is about to appear wearing a completely different costume.

Chapter 2

The alignment problem, and CTC's answer

Practice problems

Problem 2.1. Apply the collapse map: what are $\mathcal{B}(\text{AA}\varepsilon\text{B})$, $\mathcal{B}(\varepsilon\text{ABB})$, and $\mathcal{B}(\text{ABBA})$?

Solution. $\mathcal{B}(\text{AA}\varepsilon\text{B})$: merge the AA, drop the blank → AB. $\mathcal{B}(\varepsilon\text{ABB})$: nothing adjacent merges except BB, drop the blank → AB. $\mathcal{B}(\text{ABBA})$: merge BB → ABA — the trailing A is a *new* A, separated by the B, so it survives: ABA, not AB.

Problem 2.2. Show by example that the collapse order matters: find a path that yields a double letter under merge-then-drop but loses it under drop-then-merge.

Solution. Take LεL. Merge-then-drop: no adjacent repeats to merge, drop the blank → LL — the double letter survives. Drop-then-merge: dropping the blank first gives LL adjacent, which then merges → L. Only merge-then-drop lets the blank protect a genuine double letter, which is why the map is defined in that order.

Problem 2.3. Over $T = 3$ frames with alphabet $\{A, \varepsilon\}$, how many raw paths are there, and how many collapse to the transcript "A"?

Solution. Raw: $2^3 = 8$. Collapsing to A means the path shows one contiguous block of A's and blanks elsewhere: the block can occupy frames $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}$ — 6 paths. The trellis agrees: unitweight columns on the 3 states $(\varepsilon, A, \varepsilon)$ run $(1,1,0) \to (1,2,1) \to (1,3,3)$, and the two accepting states give $3+3=6$.

Problem 2.4. Using the repeat-free count $\binom{T+U}{T-U}$, how many paths spell a 2-letter transcript with no repeated letter at $T=5$?

Solution. $\binom{5+2}{5-2} = \binom{7}{3} = 35$ paths — the count 35 the road meets again when blank dominance is tallied at $T=5$.

Problem 2.5. [stretch] The chapter claims greedy decoding can be wrong. Over the two-symbol alphabet $\{A, \varepsilon\}$ at $T=2$, construct per-frame scores where the frame-wise argmax path collapses to a transcript that is NOT the highest-probability transcript.

Solution. Score both frames identically: $y_t(A) = 0.4$, $y_t(\varepsilon) = 0.6$. Greedy takes the blank at each frame — path $\varepsilon\varepsilon$ — and decodes the *empty* transcript, whose only path has mass $0.6 \times 0.6 = 0.36$. The transcript "A" pools three paths: $\mathcal{B}^{-1}(\mathrm{A}) = \{\mathrm{AA},\ \mathrm{A}\varepsilon,\ \varepsilon\mathrm{A}\}$ with mass $0.16+0.24+0.24=0.64$. (The two transcripts exhaust the outcomes: $0.36+0.64=1$.) Greedy answers "nothing was said"; the sum answers "A", by nearly two to one — frame-wise favourites lose to a transcript that spreads its weight across many paths, which is exactly why the loss is built on the sum.